Exp2:numerical simulation and performance analysis of microwave transmission line and two port networks using scattering parameters

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clc;
clear;
close all;
% ECA13 - Experiment 2
% Microwave Transmission Line and Two-Port S-Parameters
 c = 3e8;
 Z0 = 50;
L = 50e-3;
 f0 = 10e9;
% Speed of light (m/s)
% Characteristic impedance (ohms)
% Transmission line length (m)
% Center frequency (Hz)
 % Frequency sweep: 8 GHz to 12 GHz in 10 MHz steps
 f = (8e9:10e6:12e9);
N = length(f);
% Initialize S-parameters
 S11 = zeros(1,N);
 S22 = zeros(1,N);
S21 = zeros(1,N); S12 = zeros(1,N);
 % Ideal matched, lossless transmission line model
 beta = 2*pi*f/c;
 S21 = exp(-1j*beta*L);
 S12 = exp(-1j*beta*L);
 % Convert S-parameters to dB for plotting
S11_dB = 20*log10(max(abs(S11),1e-6));
 S22_dB = 20*log10(max(abs(S22),1e-6));
S21_dB = 20*log10(abs(S21));
 S12_dB = 20*log10(abs(S12));
 % Find index corresponding to 10 GHz
 [~,idx] = min(abs(f-f0));
 % Calculate performance at 10 GHz
 RL = -20*log10(max(abs(S11(idx)),1e-12));
IL = -20*log10(abs(S21(idx)));
 efficiency = abs(S21(idx))^2*100;
fprintf('Center Frequency = %.2f GHz\n',f(idx)/1e9);
fprintf('Characteristic Impedance = %.2f ohms\n',Z0);
 fprintf('Line Length = %.2f mm\n',L*1000);
 fprintf('S11 = %.4f\n',abs(S11(idx)));
fprintf('S21 = %.4f\n',abs(S21(idx)));
 fprintf('S12 = %.4f\n',abs(S12(idx)));
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fprintf('S22 = %.4f\n',abs(S22(idx)));
 fprintf('Return Loss = %.2f dB (ideal matched)\n',RL);
fprintf('Insertion Loss = %.2f dB\n',IL);
 fprintf('Transmission Efficiency = %.2f %%\n',efficiency);
% Plot S-parameter magnitudes figure;
 plot(f/1e9,S11_dB,'r','LineWidth',2);
 hold on;
 plot(f/1e9,S21_dB,'b','LineWidth',2);
plot(f/1e9,S12_dB,'g--','LineWidth',2);
plot(f/1e9,S22_dB,'m--','LineWidth',2);
 xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
 title('S-Parameter Response of a 50 Ohm Two-Port Network');
legend('S11','S21','S12','S22','Location','best');
 grid on;
 ylim([-130 5]);
hold off;
% Plot transmission phase figure;
 plot(f/1e9,unwrap(angle(S21))*180/pi,'LineWidth',2);
xlabel('Frequency (GHz)');
ylabel('S21 Phase (degrees)');
title('Transmission Phase versus Frequency');
grid on;
```

Output;

